

Experiment:- Study of LAN Topology, Network devices and Address configuration.

Objective:-

- 1> To study the LAN topology used in college campus.
- 2> To identify networking device & understand their function.
- 3> To analyse wired & wireless communication method used in the network.
- 4> To measure speed of network for wired & wireless connections.
- 5> To obtain MAC, IP Address & subnet mask.

Problem Statement - 1

- Draw & identify LAN Topology of college campus and clearly label all components such as router, switches & etc.

Soln:-

The college campus network observed this study consisted of:-

- * Layer-3 core switches in the server room.
- * Managed by POE + switches.
- * Wireless Access Point (APs)
- * Ethernet Port in classroom & laboratories

Network Topology

- Hybrid Hierarchical Topology which combines:
- 1) Tree Topology
 - 2) Extended star Topology
 - 3) Ring Topology (Server Room)

The network hierarchical layer

- ① Core Layer
- ② Distribution Layer
- ③ Access Layer

Core Layer

* Core layer contains Server Room and Server room and Server room have firewall, Router, 2-L3+ Switches, Patch Rack, Server, inverter etc.

Distribution Layer

* Distribution layer contains 4 - Switches.

* The 4 switches are →

- ① Main Build → R7 (Room no-213, 2nd floor)
- ② Main Build → R1 (Room no-301, 3rd floor)
- ③ Main Build → R6 (Room no-113, 1st floor)
- ④ Main Build → R2 (Room no-201, 2nd floor)

* This 4 - switches contains →

- ① 18 Port Gigabit Easy Smart Switch with
16 - Port

TL-SG1218MPE

SG 3428 XMP → 24 Port Gigabit & 4 Port 10 GbE SFP+, L2+ Managed Switches with 24 Port PoE+.

→ Now Let's discuss these 2 above components

① TP-link TL-SG1218MPE

- likely used for: CCTV camera, Wifi access Point, etc.

- 16 Port Support PoE+

② TP-link Omada SG3428 XMP ...

- Higher-end managed Layer 2+ Switch.
- 24 PoE+ Port, Uplinks, 10 Gb SFP
- Usually acts as a main distribution for each floor.

→ Access Layer

- Access layer contain 3 switches i.e

1) Main Build R1 - [Room no - 301, 3rd floor]

2) Main Build R4 - [Room no - 05, Ground floor]

3) Main Build R5 - [Room no - 018, Ground floor]

- Every switches acts as access layer which helps to provide network connectivity to all classroom, lab, faculty room etc.

- Access Layer directly connect

① Ethernet Port

② Wireless AP

③ Lab System

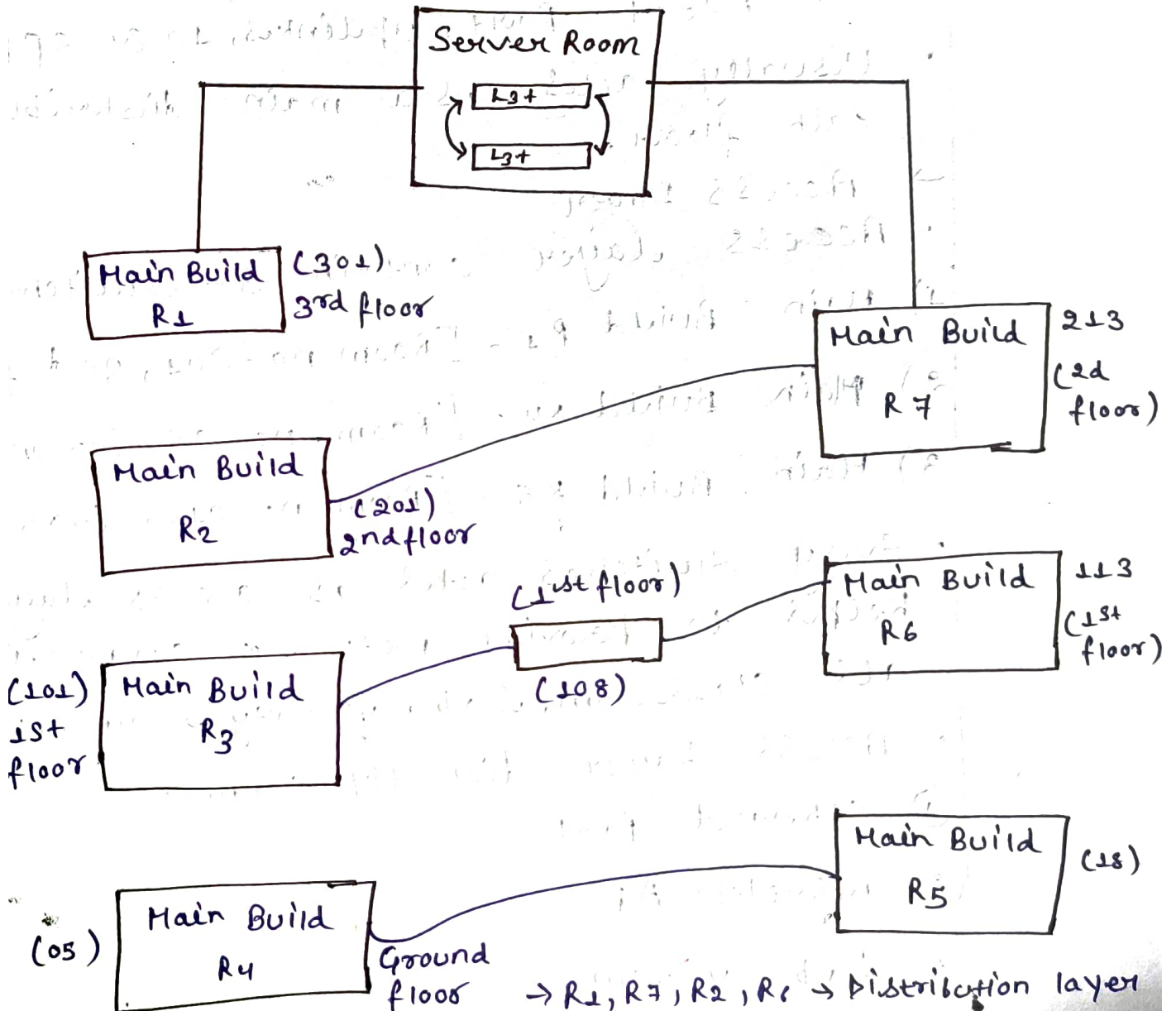
④ outdoor Wireless AP (ex-018)

Router

L3 + Switch



Topology :-

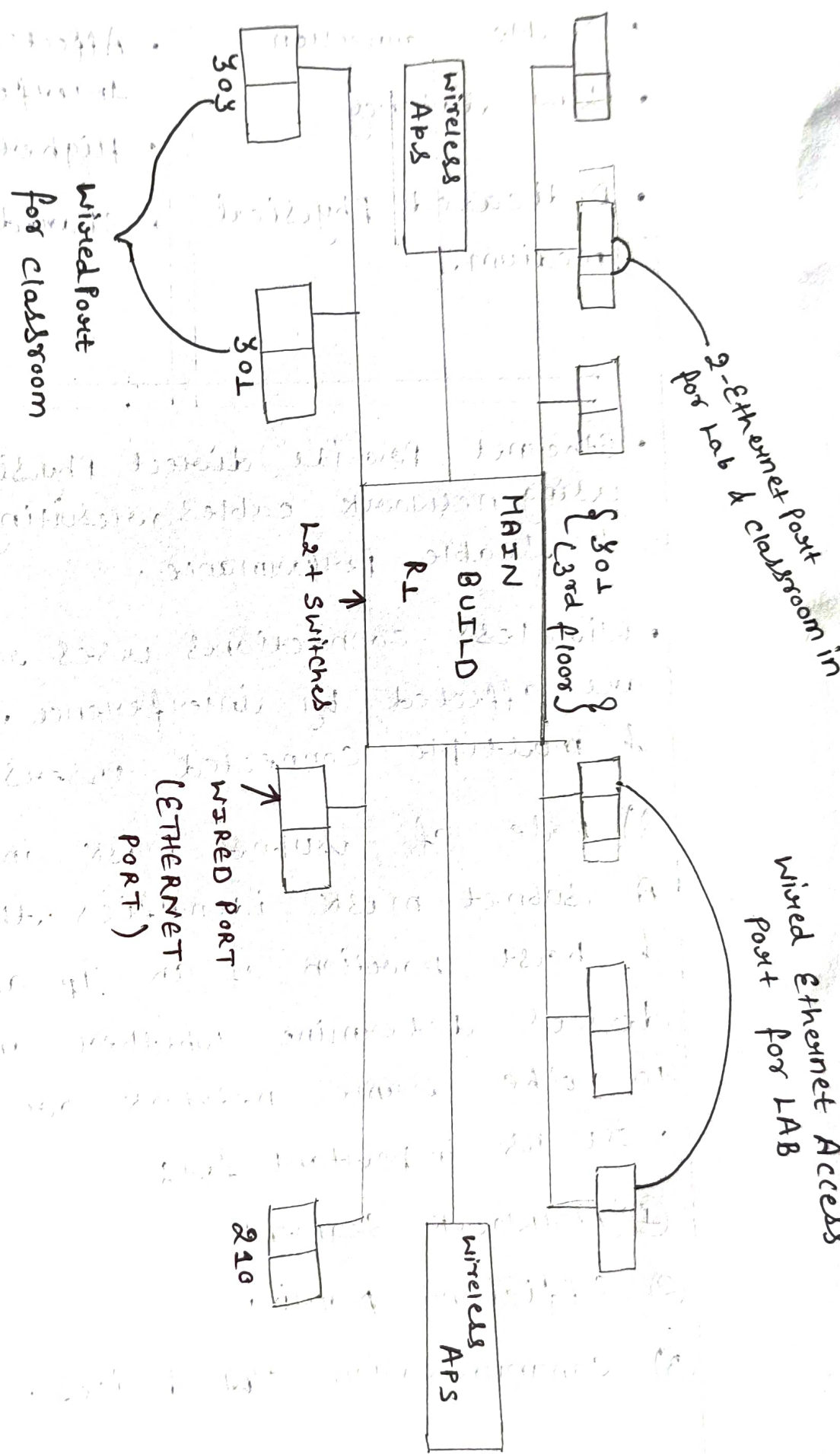


→ R1, R7, R2, R6 → Distribution layer

→ Server Room → Core Layer

→ R1, R2, R3, R4, R5, R6, R7 → Access layer

Layer Switch (3rd floor - 301)



Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\aparn> netsh wlan

The following commands are available:

Commands in this context:

?	- Displays a list of commands.
add	- Adds a configuration entry to a table.
connect	- Connects to a wireless network.
delete	- Deletes a configuration entry from a table.
disconnect	- Disconnects from a wireless network.
dump	- Displays a configuration script.
export	- Saves WLAN profiles to XML files.
help	- Displays a list of commands.
IMV	- Commands for IMV logging.
refresh	- Refresh hosted network settings.
reportissues	- Generate WLAN smart trace report.
set	- Sets configuration information.
show	- Displays information.
start	- Start hosted network.
stop	- Stop hosted network.

To view help for a command, type the command, followed by a space, and then type ?

type ?.

PS C:\Users\aparn> netsh wlan show interfaces

There is 1 interface on the system:

Name	: Wi-Fi
Description	: Qualcomm Atheros QCA9377 Wireless
GUID	: 1ad1633b-f00a-4160-a5cb-640204e9d
Physical address	: d8:c0:a6:ee:51:05
Interface type	: Primary
State	: disconnected
Radio status	: Hardware On Software On

```
PS C:\Users\aparn> ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet:

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::ddad:20c:30eb:b3cd%4  
IPv4 Address. . . . . : 192.168.56.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
PS C:\Users\aparn> getmac
```

PS C:\Users\aparn> getmac

Physical Address	Transport Name
D8-C0-A6-EE-51-05	Media disconnected
D8-C0-A6-EE-51-04	Media disconnected
0A-00-27-00-00-04	\Device\Tcpip_{157FC536-B55C-4A52-9EF9-1EF77AAAA4A4}

PS C:\Users\aparn> wmic
wmic:root\cli>

Difference b/w wired & wireless speed?

<u>Wired Network</u>	<u>Wireless Network</u>
• faster communication • Stable connection • Low latency • Dedicated physical medium.	• Slower communication • Affected by distance / interference • Higher latency • Shared radio medium.

- Ethernet provide direct physical communication using network cables, resulting in higher speed & stable performance.
- Wireless connections uses radio waves which are affected by interference, distance obstacle & multiple connected users.

Q) Role of subnet mask in network?

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belong to the same network or different network.

• It is important for:

- ① Network Segment
- ② Efficient Routing
- ③ Communication b/w devices.

OBSERVATION

During the study of the college network infrastructure:-

- ① A centralized server was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-link PoE + switches were used in network cabinets.

Advantage of Existing Network

- Scalable architecture.
- Centralized management.
- Supports PoE + device.
- Easy expansion of floors & classroom.
- Support both wireless & wired.
- Redundant link for failure.

CONCLUSION

The observed campus LAN follows a hybrid hierarchical topology consist of Tree, Extended star, Ring. The network uses managed PoE switches, wireless access point, Ethernet port & centralized server inf. to provide efficient wired & wireless communication across the campus.